

Intelligent travelling:

Smarter public transport (Know everything): We've all been in that situation where we had to wait for a bus at the bus stop for a long time in the heat. Even though we can see the bus timings on the public transport portal online for each individual bus, we know that owing to traffic snarls and punctuality issues, the buses are never on time. IoT can come to the rescue here by making use of wireless technology in buses and bus stops. A simple system can be devised which shares the current location of the incoming bus from the road, difference in distance between the bus and the next bus stop. Citizens who want to hop onto a bus from their nearest bus stop can use their smartphones to get notified of the bus' location and leave accordingly. All of the above and then some more can be seen in action with something called as Micro-Navigation for urban bus passengers using the Internet of Things which is being put to test in several countries. The current public transport system can be quite confusing and difficult to navigate for passengers such as disabled people, senior citizens and tourists. This is why the Urban Bus Navigator (UBN) which was successfully deployed and tested in Madrid, Spain should be the next big thing in urban bus transportation. UBN employs a novel concept that uses a context-aware navigation system that supports micro and macro navigation with continuous dynamic trip tracking, semantic bus ride detection and dynamic navigation information. The system can monitor the ongoing trip of the passenger and offer a plan for going ahead in the trip and tell the passengers when should they get off, is the next bus in their trip is bound to reach a bus stop that's the same for the current bus on the same route? And more. The knowing everything aspect of UBN is that a passenger is always provided with the dynamic information that he/she will ever need. If for example a passenger takes a wrong bus, gets off at the wrong bus-stop, the system immediately raises a flag and notifies the passenger. It also charts out an alternate plan with a new set of instructions to navigate accordingly. It works almost like how a simple Google Maps navigation works but is more centric to the passenger on the bus with the bus providing connectivity using WiFi and GPS and is connected to a central system. You can read more about UBN in this research paper: <http://dgit.in/UrbanBusSys>

Better on-road Safety: Safety on the road is of utmost importance and so plenty of new ideas for improving road safety are being adopted by governments of developed countries. One such example is Volvo Cloud which was demonstrated at the Mobile World Congress in Barcelona. Volvo